

PARALLEL REACTIVE CIRCUITS

AC CIRCUITS

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TOPIC OUTLINE

Phasor Diagram

Parallel R-L Circuit

Parallel R-C Circuit

Parallel R-L-C Circuit

Power Factor



PHASOR DIAGRAM



R-X-Z COMPLEMENT

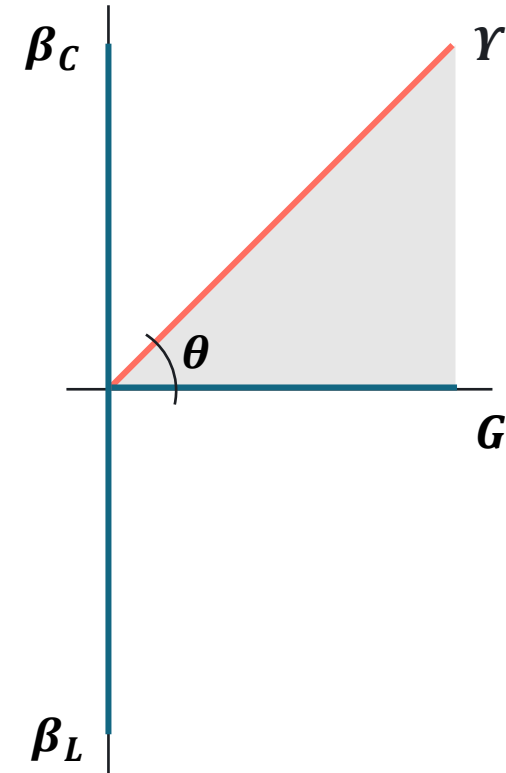
Ohms (Ω)	Siemens (S)
Resistance R	Conductance $G = 1/R$
Reactance X	Susceptance $\beta = 1/X$
Impedance Z	Admittance $Y = 1/Z$



PHASOR DIAGRAM

Phasor diagram is a graphical representation of magnitude and phase relationship between sinusoidal quantities.

Phasor Diagram



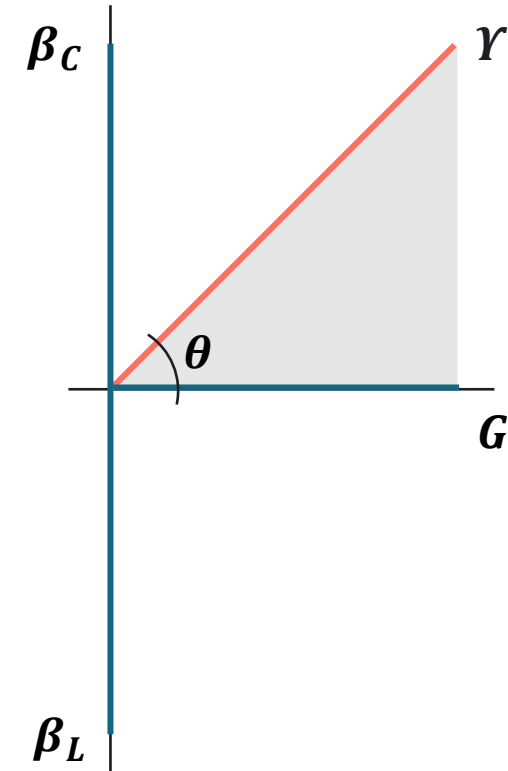
ADMITTANCE

The admittance (Y) represents the total ability offered by circuit elements (including resistance and reactance) to conduct alternating current (AC).

Formula

$$Y = \frac{1}{Z}$$

Phasor Diagram



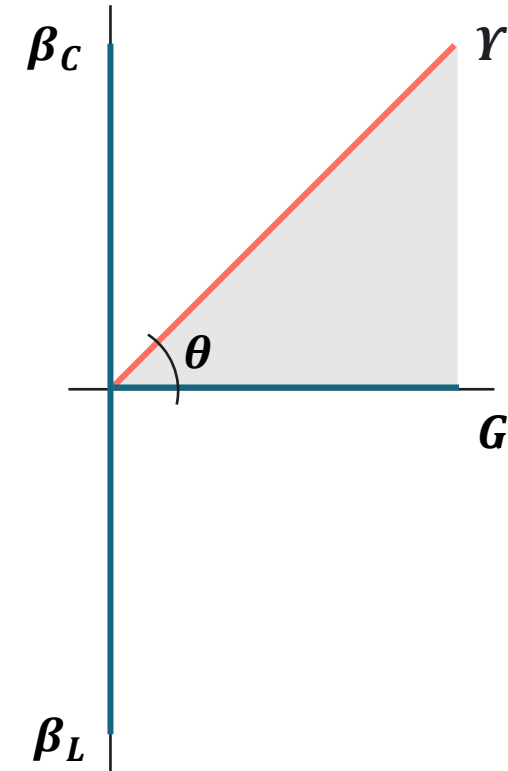
CAPACITIVE SUSCEPTANCE

The capacitive susceptance (β_C) represents the ability of capacitor to allow the flow of alternating current (AC).

Formula

$$\beta_C = \frac{1}{X_C}$$

Phasor Diagram



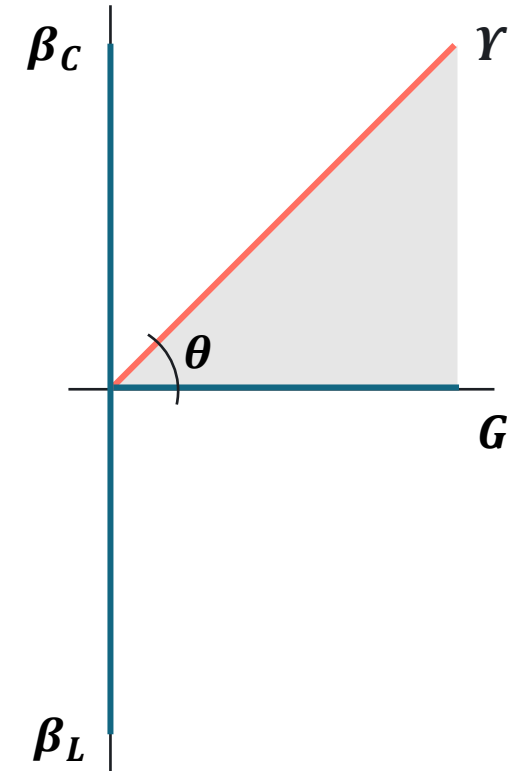
INDUCTIVE SUSCEPTANCE

The inductive susceptance (β_L) represents the ability of inductor to allow the flow of alternating current (AC).

Formula

$$\beta_L = \frac{1}{X_L}$$

Phasor Diagram

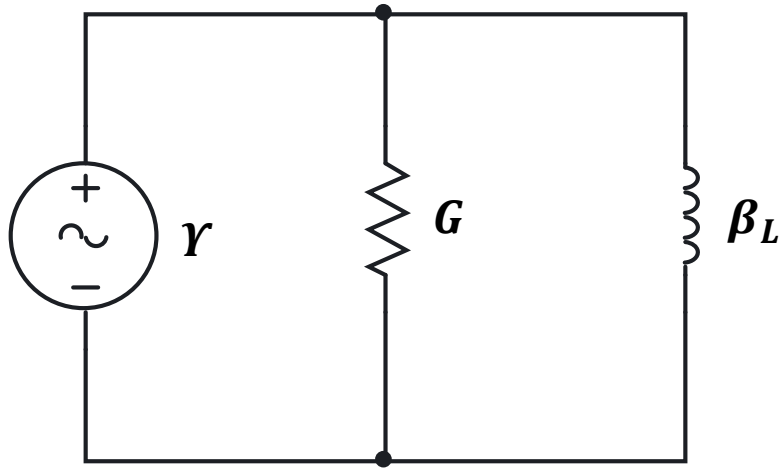


PARALLEL R-L CIRCUIT



PARALLEL R-L CIRCUIT

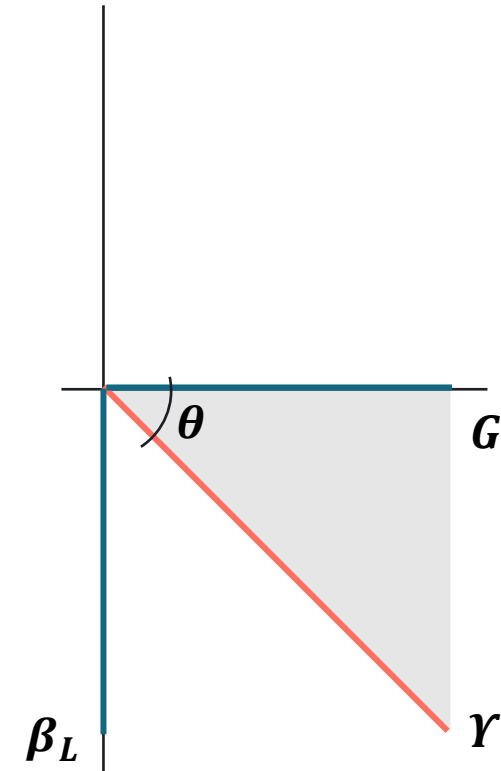
Circuit Diagram



Formula

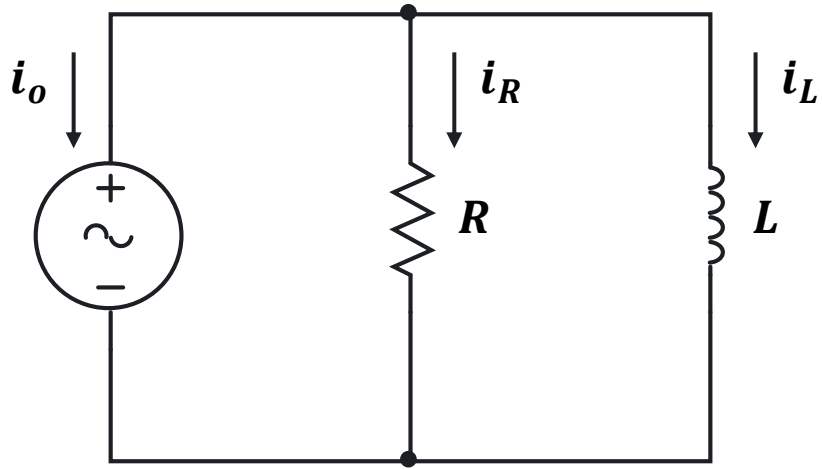
$$\gamma = G + j\beta_L$$

Admittance Phasor Diagram



PARALLEL R-L CIRCUIT

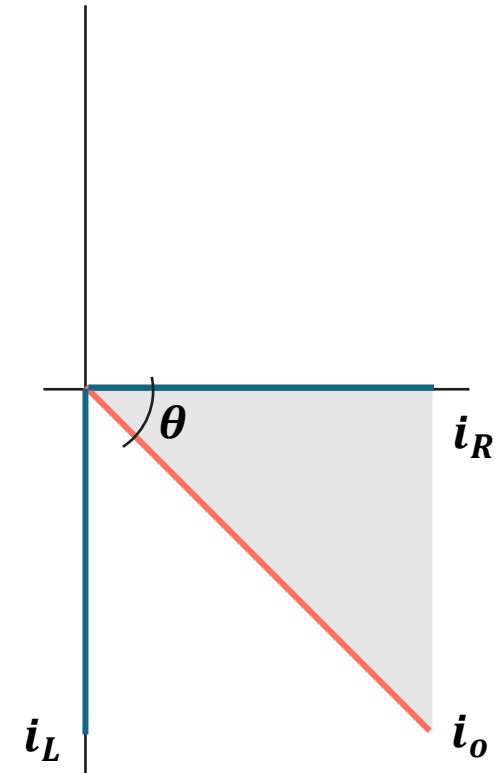
Circuit Diagram



Formula

$$i_o = i_R + j i_L$$

Current Phasor Diagram



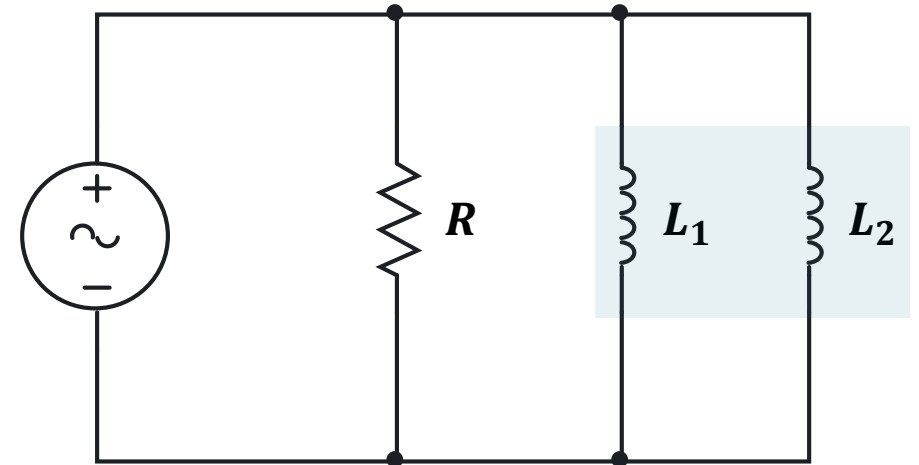
TOTAL INDUCTANCE

In a parallel circuit, the total inductance (L_o) is analogous to the total resistance in parallel circuit.

Formula

$$\frac{1}{L_o} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_n}$$

Parallel network



EXERCISE

A 120V, 60Hz source is connected to a parallel combination of 30Ω resistor and 100mH inductor.

Solution

Determine the following:

- a. Equivalent impedance ($\mathbf{Z}$)
- b. Total current ($\mathbf{i_o}$)
- c. Current through the resistor ($\mathbf{i_R}$)
- d. Current through the inductor ($\mathbf{i_L}$)

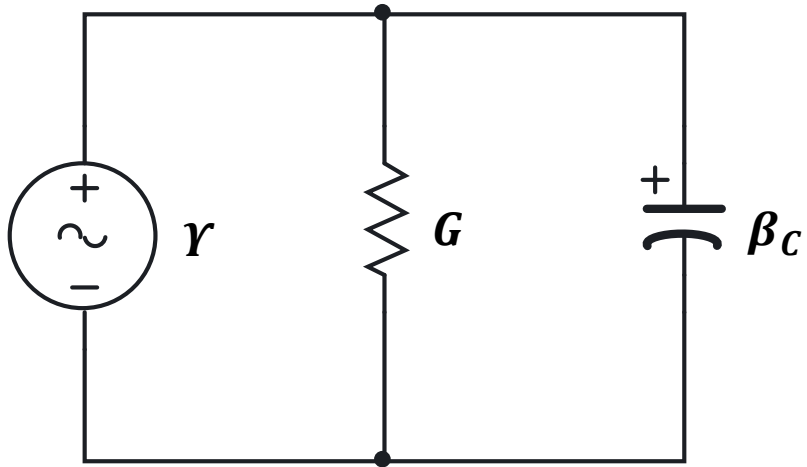


PARALLEL R-C CIRCUIT



PARALLEL R-C CIRCUIT

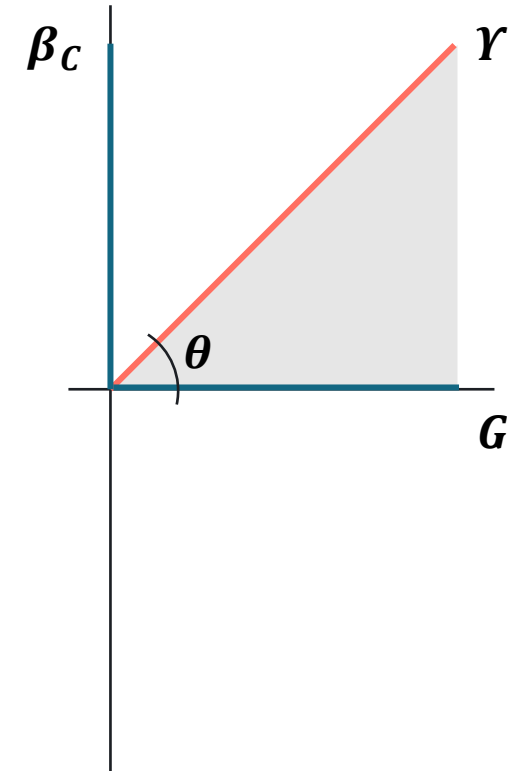
Circuit Diagram



Formula

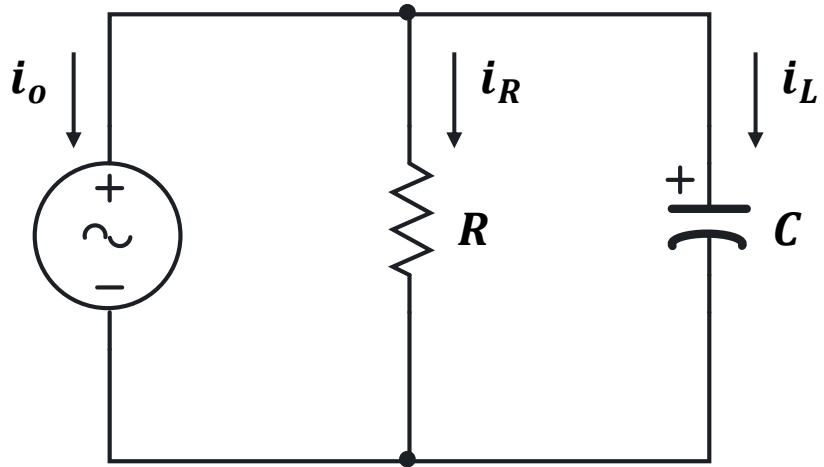
$$\gamma = G + j\beta_c$$

Admittance Phasor Diagram



PARALLEL R-C CIRCUIT

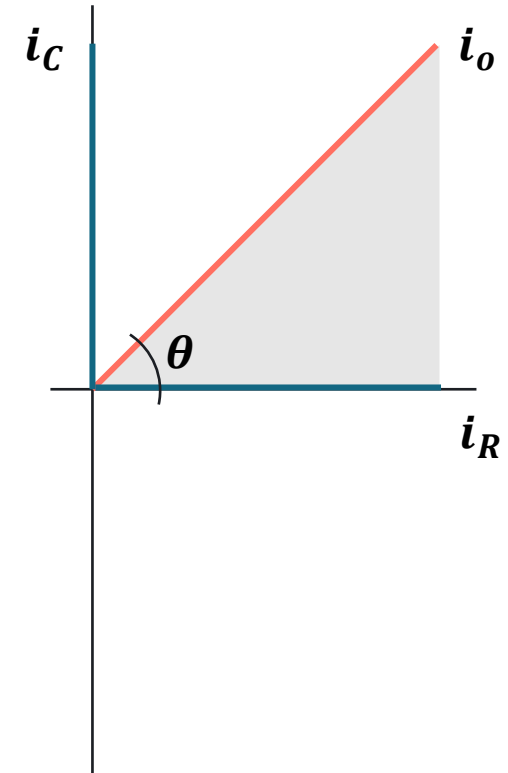
Circuit Diagram



Formula

$$i_o = i_R + j i_C$$

Current Phasor Diagram



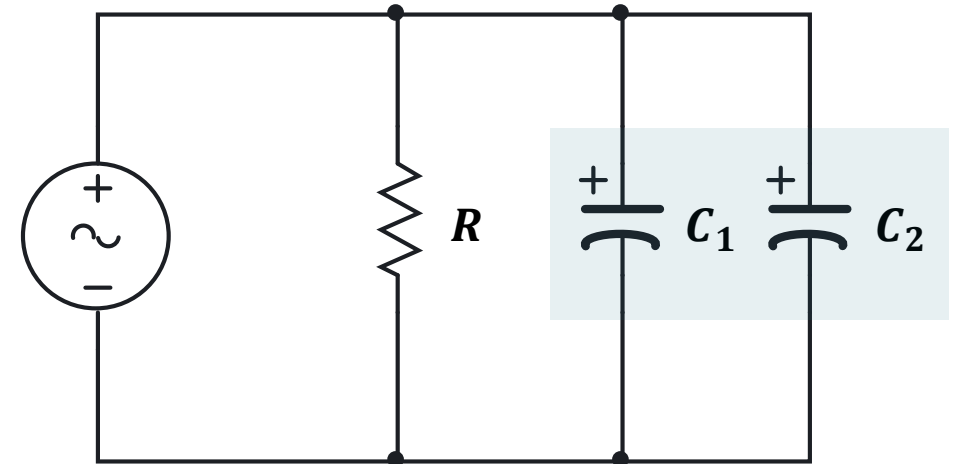
TOTAL CAPACITANCE

In a parallel circuit, the total capacitance (C_o) is the sum of all individual capacitances.

Formula

$$C_o = C_1 + C_2 + C_3 \dots C_n$$

Series network



EXERCISE

A 120V, 60Hz source is connected to a parallel combination of 10Ω and a $120\mu\text{F}$ capacitor.

Solution

Determine the following:

- a. Equivalent impedance ($\mathbf{Z}$)
- b. Total current ($\mathbf{i_o}$)
- c. Current through the resistor ($\mathbf{i_R}$)
- d. Current through the capacitor ($\mathbf{i_C}$)

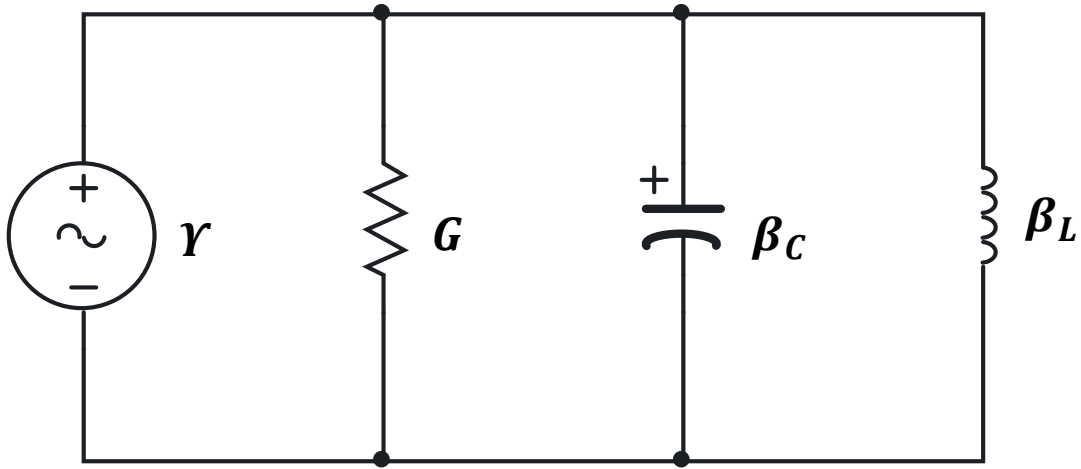


PARALLEL R-L-C CIRCUIT



PARALLEL R-L-C CIRCUIT

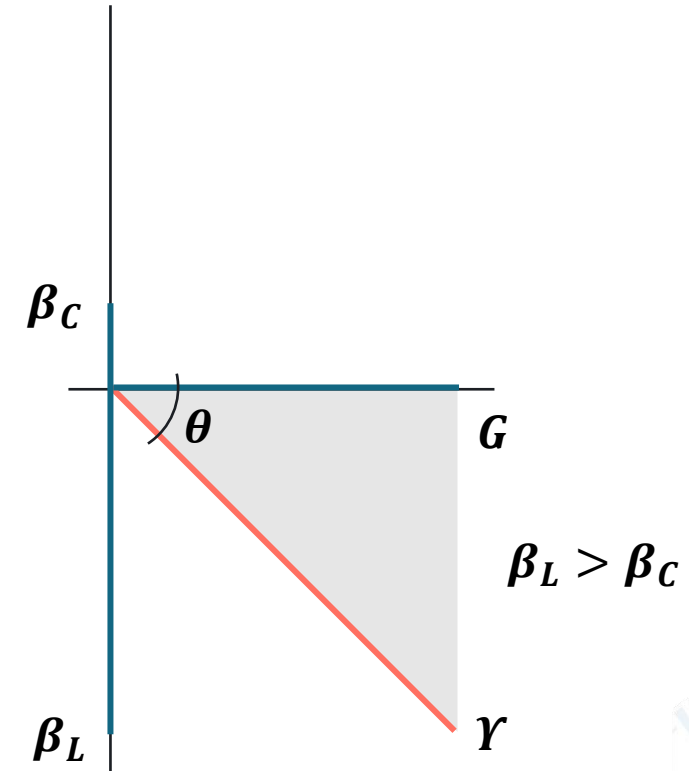
Circuit Diagram



Formula

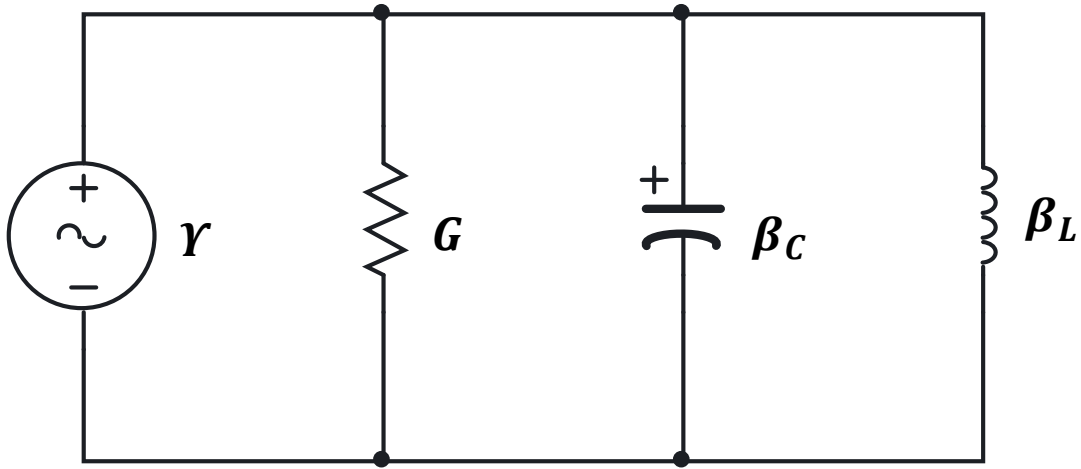
$$Y = G + j(\beta_C - \beta_L)$$

Admittance Phasor Diagram



PARALLEL R-L-C CIRCUIT

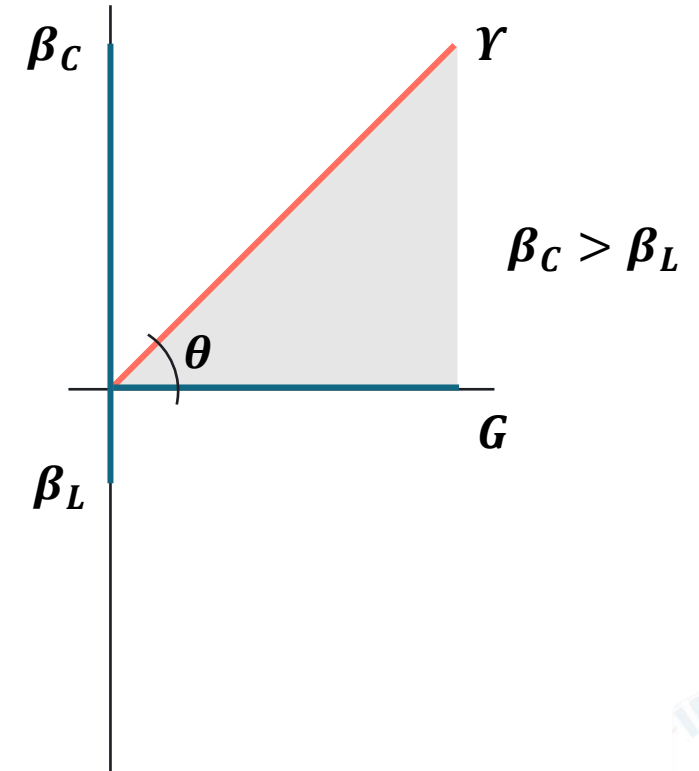
Circuit Diagram



Formula

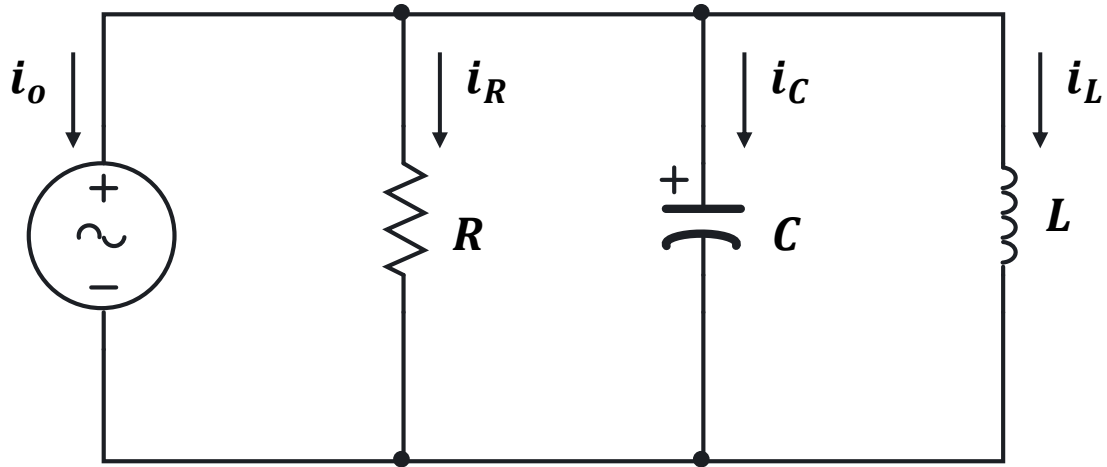
$$Y = G + j(\beta_C - \beta_L)$$

Admittance Phasor Diagram



PARALLEL R-L-C CIRCUIT

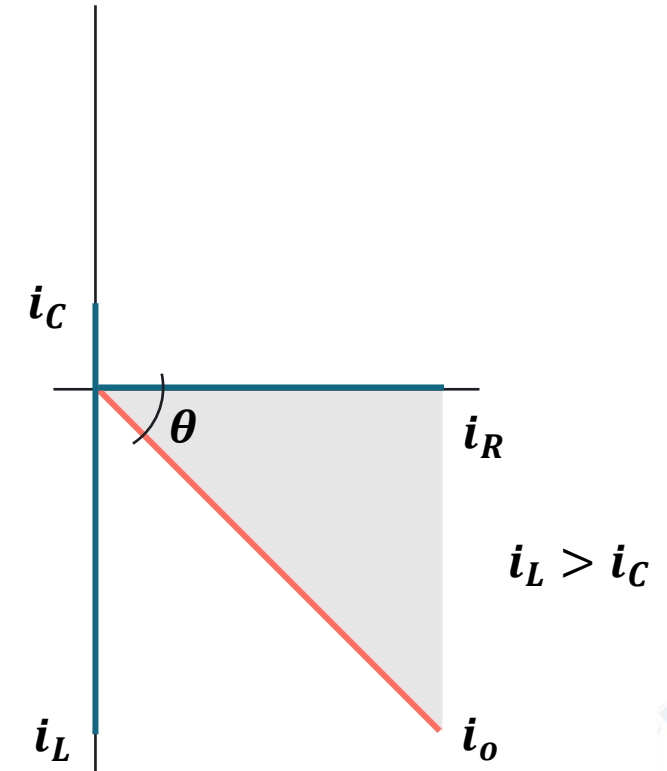
Circuit Diagram



Formula

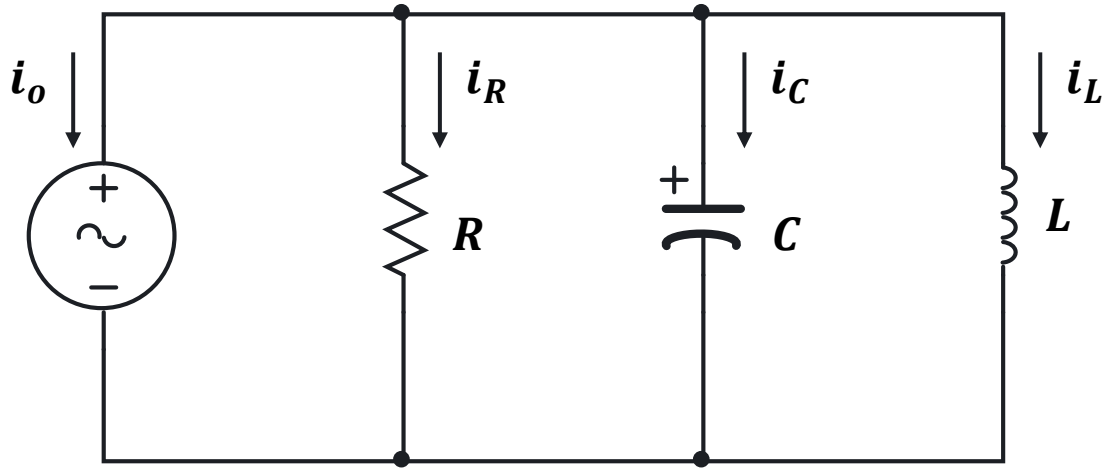
$$i_o = i_R + j(i_C - i_L)$$

Current Phasor Diagram



PARALLEL R-L-C CIRCUIT

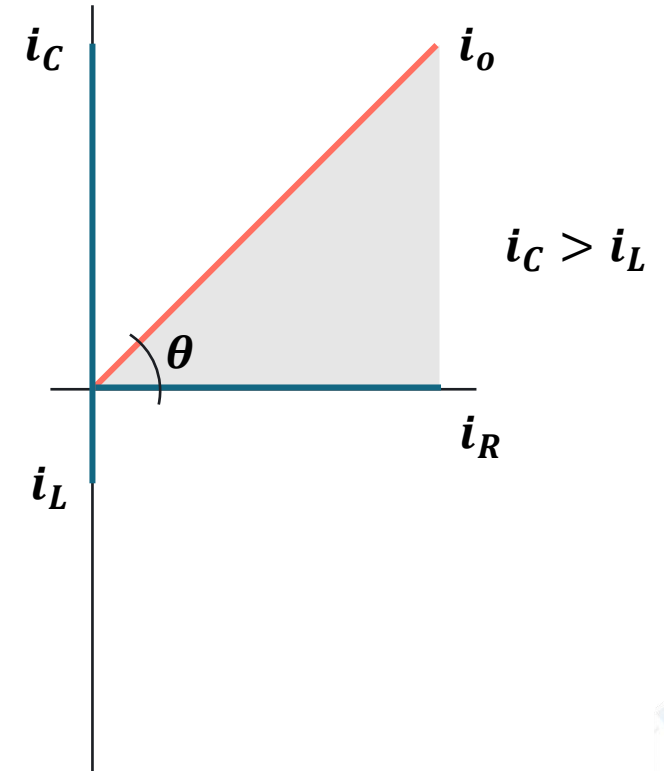
Circuit Diagram



Formula

$$i_o = i_R + j(i_C - i_L)$$

Current Phasor Diagram



EXERCISE

A 50Ω resistor, a 20mH coil and a $5\mu\text{F}$ capacitor are all connected in parallel across a 50V, 100Hz supply.

Solution

Determine the following:

- a. Equivalent impedance ($\mathbf{Z}$)
- b. Total current ($\mathbf{i_o}$)
- c. Current through the resistor ($\mathbf{i_R}$)
- d. Current through the inductor ($\mathbf{i_L}$)
- e. Current through the capacitor ($\mathbf{i_C}$)



POWER FACTOR



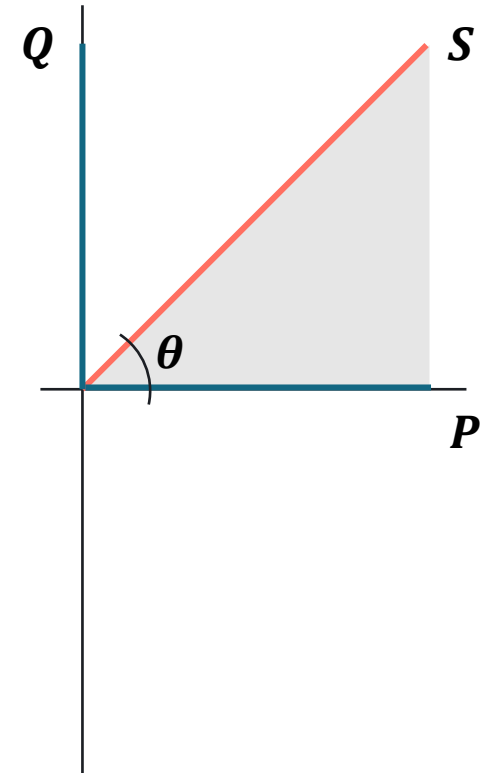
POWER FACTOR

Power Triangle

The power factor ($\cos \theta$) represents the ratio of true power to apparent power.

Formula

$$\cos \theta = \frac{P}{S}$$



TRUE POWER

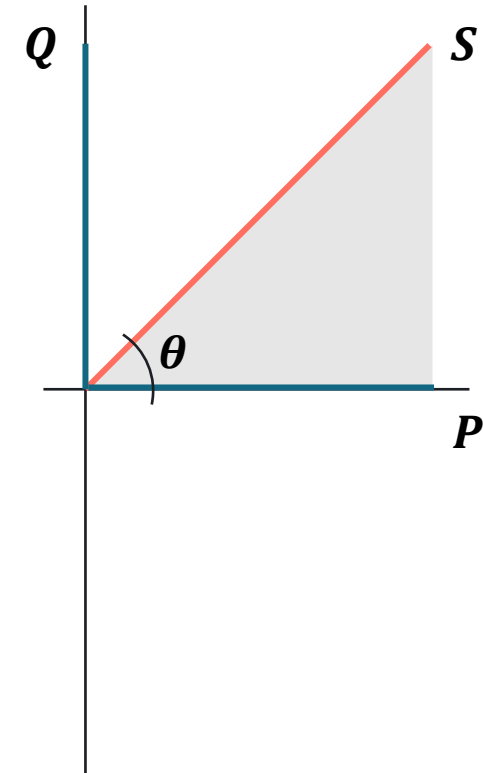
The true power (P) is the actual power consumed by resistive components of a circuit.

Formula

$$P = vi \cos \theta$$

unit: Watt (W)

Power Triangle



REACTIVE POWER

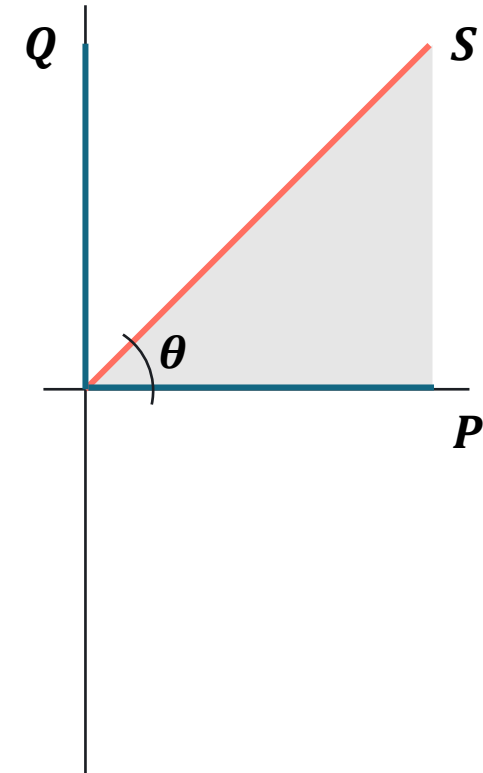
Power Triangle

The reactive power (Q) is the power consumed by inductive or capacitive components of a circuit.

Formula

$$Q = vi \sin \theta$$

unit: Volt-Ampere Reactive (VAR)



APPARENT POWER

The apparent power (S) is the vector sum of true power and reactive power, representing the total power supplied by the source.

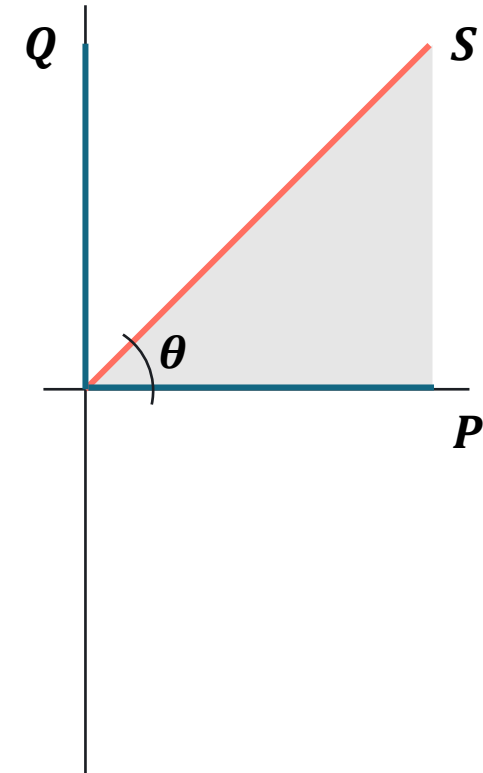
Formula

$$S = P + jQ$$

$$S = vi$$

unit: Volt-Ampere (VA)

Power Triangle



EXERCISE

A 50Ω resistor, a 20mH coil and a $5\mu\text{F}$ capacitor are all connected in parallel across a 50V , 100Hz supply.

Solution

Determine the following:

- The power factor (**$\cos \theta$**)
- The true power (**P**)
- The reactive power (**Q**)
- The apparent power (**S**)



LABORATORY

